Purdue University School of Materials Engineering
Spring 2026

MSE 382 Mechanical Response of Materials

INSTRUCTOR

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TEXT

Meyers, M. A., & Chawla, K. K. (2025). *Mechanical behavior of materials* (3rd ed.). Cambridge University Press.

GOALS

This course explores the mechanical response of materials including elasticity, plasticity, structural transformations, and failure addressing both the underlying microscopic mechanisms (e.g. dislocations motion and mass diffusion) and macroscopic response. The mechanical behavior of metals, ceramics, polymers, and composites will be related to their underlying atomic structure and elasticity and plasticity concepts for single crystal, polycrystalline, and amorphous materials. Practical design consideration for deformation and failure will be introduced.

LEARNING OUTCOMES

1. Conversant with conventional nomenclature, units and notation of mechanical behavior. Examples: Stress, strain, yield strength, fracture strength; yield criteria, flow rules, creep, fracture.
2. Estimate relative ranges and values for important properties of common engineering materials. Examples: Young's modulus at room temperature for steels versus most polymers; yield strength ranges for ductile metals.
3. Recognize mechanisms for important mechanical behaviors. Examples: Plastic deformation by dislocation glide; plastic deformation by molecular rearrangement; dilatant deformation processes; strengthening mechanisms in crystalline solids.
4. Predict mechanical responses based on given information. Examples: Will yielding occur given a stress state and properties? Will fracture occur given a stress state and properties? How will the material change shape for the applied deformation? Which slip system(s) will be operative?
5. Explain the fundamental basis for important deformation mechanisms or processes. Examples: Dislocation glide, cross-slip, dislocation climb, precipitation strengthening; molecular orientation; Nabarro-Herring creep; plane strain fracture.
6. Apply the mechanisms of deformation to component design or alloy design.

GRADING

Weekly Quizzes (Friday)	60%
Comprehensive Final exam	30%
Homework	10%

Grading will be done using a 10-point scale such that 100-97= A+, 96-93=A, 92-90=A- and so on.

Homework & Quizzes will be based on the lecture and reading assignments and will be given weekly. Quizzes will typically be given on Friday. The lowest homework and quiz grades will be dropped.

COURSE SCHEDULE

Week	Topics
1	Introduction – Chapter 1
2	Elasticity – Chapter 2
3	Anisotropy and tensor formulation – Chapter 2
4	Plastic deformation – Chapter 3
5	Plasticity of polymers and glasses; yield criteria – Chapter 3
6	Dislocations – Chapters 4 & 5 (selected)
7	Dislocations – Chapters 4 & 5 (selected)
8	Strengthening mechanisms – Chapters 6 & 10 (selected)
9	Strengthening mechanisms – Chapters 6 & 10 (selected)
10	Spring Break
11	High temperature and rate dependent deformation – Chapter 13
12	High temperature and rate dependent deformation – Chapter 13
13	Fracture mechanics – Chapters 7,8 & 9 (selected)
14	Fracture mechanics – Chapters 7,8 & 9 (selected)
15	Fatigue of materials – Chapter 14
16	Review
17	Final Exam

PROCEDURES

Lectures will emphasize the more important points, but you are also responsible for the details in the reading that time may not permit covering in lecture.

Homework

Homework assignments and solutions will be posted on Blackboard.

Quizzes and Examinations are closed books. Calculators are allowed. Your lowest quiz score will be dropped and not counted in the final average.

Make-up exams or quizzes will be given only for the following verifiable reasons: serious illness, family emergencies, or official university absence.

Final grades

Your recorded grades on homework, quizzes, and exams will be available on Blackboard.

GENERAL POLICIES

Students are responsible for knowing the information contained in this syllabus. It is expected that students will attend and actively participate in all course lectures. All individuals will always treat each other with respect and courtesy. Students should minimize disruptive behavior while in lecture, such as excessive talking, laughing, texting, and web-surfing. All course announcements will be made in lecture and posted as "Announcements" on Brightspace. Lastly, the course instructor retain the right to ask disruptive or disrespectful students to leave the classroom or lab at any point.

ACADEMIC INTEGRITY

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Purdue University Regulations, Part 5, Section III- B-2-a describes the formal policies governing academic dishonesty. A guide providing specific examples, tips, and consequences is available at: http://www.purdue.edu/odos/osrr/academicintegritybrochure.php . These rules cover all graded assignments, including lab reports, quizzes, and exams. Copying or sharing any part of the graded assignment from any source is a clear violation of university policy. Transcribing, paraphrasing or following the outline of someone else's solution(s) is cheating. The test is simple: If you write down your solution after reading or hearing someone else's solution, you are cheating. Likewise, if you share or exchange your solutions with another student, in written or spoken form, you are cheating. Discussion between students of the broad concepts and general approach to lab reports is encouraged, and images, data, and results will be shared between groups; however, the solutions, charts, and graphs you turn in for grading must be your own original work. Thus, sharing work beyond raw data and broad discussions will be considered plagiarism.

Students found to engage in academic dishonesty are subject to sanctions. For a first offense, the student will receive a zero score for the assignment, report, quiz, or exam. Depending on the severity of the offense, the case may be referred to the Office of the Dean of Students for further action. A second offense will result in a failing grade for the course and definite referral to the Office of the Dean of Students. Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

EMERGENCIES

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Information about changes in this course will be posted on Brightspace and/or distributed via email to your @purdue.edu email address. Emergency preparedness is your personal responsibility. Purdue University is actively preparing for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus. For full details, visit: http://www.purdue.edu/ehps/emergency_preparedness/

- For any emergency, call 911.
- There are nearly 300 Emergency Telephone Systems throughout campus that connect directly to the Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
- If we hear a fire alarm, we will immediately evacuate the building and proceed outside. Do not use the elevator.
- If we are notified of a Shelter in Place requirement for a tornado warning, we will shelter in the lowest level of this building away from windows and doors.
- If we are notified of a Shelter in Place requirement for a hazardous materials release or civil disturbance, we will shelter in our classroom shutting any open doors and windows.
- ARMS Building Emergency Plan: https://www.purdue.edu/ehps/emergency_preparedness/bep/arms- bep.html
- If you receive any Purdue ALERT via text, twitter, etc, that is relevant to you, your classmates, the building, or the surrounding area, please ensure Prof. Wharry is aware of the alert message and take appropriate actions.

USE OF COPYRIGHTED MATERIALS

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally. Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor.

MENTAL HEALTH & WELLNESS

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact

Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

STUDENTS WITH DISABILITIES

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247.

NONDISCRIMINATION

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services and activities consistent with applicable federal, state and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Executive Memorandum No. D-1, which provides specific contractual rights and remedies. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

Community, Assistance and Resources for Engineering Students

The Community, Assistance and Resources for Engineering Students (CARES) Hub is available to support the well-being of all engineering students. The CARES Hub, located in ARMS 1261 and 1264, is a welcoming, inclusive space for students to study, connect, grab a snack, and relax during the week. CARES also offers wellness activities, educational workshops, peer mentoring, social events, and on-site counseling. If you need help handling stress or working through a problem, schedule time with our CARES Hub therapist. Please visit the CARES Hub webpage to learn more. The CARES staff look forward to communicating with you!

Guidance on Generative AI

As a general policy, I would recommend following the AI policy from Purdue's Data Mine project as summarized below:

(https://the-examples-book.com/crp/students/fall2023/syllabus#guidance-on-generative-ai) "As the world of machine learning, deep learning, and AI continues to evolve we wanted to offer some guidance on The Data Mine's perspective for generative AI tools, such as ChatGPT. New emergent technologies can be incredibly valuable tools. However, at the same time it's important to keep perspective on how and when we utilize these new systems.

When using ChatGPT (or other generative AI) on a Data Mine project:

Never share a company's code, data, information, or any other proprietary property with the tool.

While not all tools incorporate user input into their training, it's a very common practice and can lead to breaches in the NDA agreements.

Always question the response that the tool provides.

It's OK to ask different apps for suggestions on things like common algorithms or good starting points for problem solutions. However, it's VITAL to understand factors like where the solutions fit, how they perform, and how to measure their performance.

It's OK for a tool to recommend an algorithm for research. It's unacceptable to assume that the algorithm is the only correct answer and to not be able to explain why it was chosen. (ChatGPT told me won't be accepted.)

It's also occasionally possible that the tool will make up an answer, and you don't want to get stuck presenting false information.

If you're ever unsure about if a tool can be used, ask your mentor and The Data Mine BEFORE you use it.

We want to use new tools and adapt to the new environments, but our number 1 priority is to provide a safe and secure data environment. We can't do anything that puts that at risk.

When using generative AI for code it's very important to understand the fundamental code's functionality.

While generative AI can easily write if/else functions or for loops, if you don't understand how they work you will have a much harder time when it comes to writing a novel or highly specific code function.

Generative AI is great to help with ideas, but shouldn't be used with no thought.

As with any new technologies, the world of generative AI is changing quickly. We encourage open discussion and welcome any feedback to The Data Mine concerning these technologies."